

Mutual Fund Analytics Platform

Capstone Project Report

End-to-End ETL, Exploratory Data Analysis, Performance & Risk Analytics on Indian Mutual Fund Data

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GitHub Repository: github.com/Tejaswini-0625/bluestock_mf_capstone

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1. Executive Summary

This report presents the findings of an end-to-end Mutual Fund Analytics Platform built during a data analytics internship at Bluestock Fintech Pvt. Ltd. The project covers the full analytics lifecycle: ETL, data cleaning, exploratory data analysis, performance analytics (returns, Alpha/Beta, Sharpe/Sortino ratios), risk analytics (Value-at-Risk, Conditional VaR), investor behaviour analysis (SIP cohort retention), and portfolio composition analysis (sector concentration via HHI), consolidated into an interactive Power BI dashboard.

The dataset spans 40 mutual fund schemes across 10 major Indian AMCs, over 64,320 NAV observations (Jan 2022 – May 2026), 32,778 investor transactions from 5,000 unique investors, and 322 portfolio holding records across 14 sectors.

Key Findings

- Total industry AUM grew from ₹30.18 lakh crore (Mar 2022) to ₹62.74 lakh crore (Dec 2025) — an annualized growth rate of approximately 21.5%.
- SBI Mutual Fund leads with 19.9% AUM market share, followed by ICICI Prudential (17.1%) and HDFC (14.8%); the fund-house HHI of 1,282 indicates a moderately concentrated but competitive market.
- Monthly SIP inflows nearly tripled between 2022 (₹1,49,437 Cr annual total) and 2025 (₹3,35,740 Cr annual total), reflecting rising retail participation via systematic investing.
- Liquid funds attracted the highest net inflows (₹4,51,275 Cr, Apr 2024–Mar 2025), followed by Sectoral/Thematic funds (₹1,03,829 Cr) — consistent with investors parking surplus cash short-term while rotating into thematic bets.
- Portfolio sector allocation is moderately concentrated (HHI = 994), led by Banking (19.3% of holding value), IT, and Pharma.
- The average fund in the study generated a 3-year annualized return of 14.09% with an average Alpha of 1.25% over its benchmark — indicating modest but consistent outperformance across the sample.
- SIP investor retention drops sharply after month 1 (from 100% to ~16%), then stabilizes near 13–17% through month 5 — a pattern consistent with a mixed cohort of long-tenure recurring SIP investors and a large share of short-lived or irregular contributors.
- A Power BI dashboard aggregation issue was identified and corrected: summing AUM across quarterly snapshots (Mar/Sep/Dec) within a year double- and triple-counted values; the correct method uses the latest snapshot per year, which was applied throughout this report.

2. Introduction

2.1 Project Background

Bluestock Fintech Pvt. Ltd. required a data-driven view of the Indian mutual fund industry to support investment research and product decisions. This capstone was scoped as a 7-day sprint to build a reusable analytics pipeline — from raw data ingestion to a decision-ready BI dashboard — using open NAV data

(mfapi.in) alongside synthetic investor transaction and portfolio holding datasets modeled on real-world structures.

2.2 Objectives

- Build a robust ETL pipeline to clean and structure fund, NAV, AUM, SIP, and investor transaction data.
- Compute industry-standard performance metrics: CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown.
- Quantify downside risk using Value-at-Risk (VaR) and Conditional VaR (CVaR) at the 95% confidence level.
- Analyze investor behaviour through SIP cohort retention and demographic segmentation.
- Assess portfolio concentration risk using the Herfindahl-Hirschman Index (HHI) across sectors and fund houses.
- Deliver a composite Fund Scorecard ranking all 40 schemes on a blended risk-return basis.
- Present all findings through an interactive Power BI dashboard and this consolidated report.

2.3 Scope of Data

Dataset	Rows	Description
Fund Master	40	Scheme metadata: AMC, category, expense ratio, fund manager, risk grade
NAV History	64,320	Daily NAV per scheme, Jan 2022 – May 2026
AUM by Fund House	90	Quarterly/annual AUM snapshots per AMC, 2022–2025
Monthly SIP Inflows	48	Industry-wide SIP metrics, Jan 2022 – Dec 2025
Category Inflows	144	Net inflow/outflow by category, Apr 2024 – Mar 2025
Industry Folio Count	21	Investor folio counts by asset class, 2022–2025
Scheme Performance	40	Returns, Alpha, Beta, Sharpe, Sortino, Drawdown per scheme
Investor Transactions	32,778	Individual transactions across 5,000 investors, 2024–2025
Portfolio Holdings	322	Stock-level holdings for all 40 schemes, Dec 2025
Benchmark Indices	8,050	Daily close values for NIFTY50 and other benchmarks

3. Methodology

3.1 Tools & Technology Stack

- Languages & Libraries: Python (Pandas, NumPy, SciPy, Matplotlib, Seaborn)

- Database: SQLite with SQLAlchemy ORM, star schema design
- Visualization & BI: Power BI (4-page interactive dashboard), Jupyter Notebooks
- Data Source: mfapi.in (live NAV data), SEBI-referenced expense ratio validation

3.2 Data Pipeline

Raw data was ingested via Python scripts, validated against SEBI expense-ratio norms, and cleaned (forward-filled NAV gaps, standardized column names, resolved Windows encoding issues). Cleaned tables were loaded into a SQLite star schema with fund_master as the dimension table and nav_history, investor_transactions, portfolio_holdings as fact tables. This structure supports efficient joins for performance and risk computation.

3.3 Analytical Methods

- CAGR & Returns: computed from NAV history over 1, 3, and 5-year rolling windows.
- Alpha/Beta: linear regression of scheme daily returns against benchmark (NIFTY100 TRI) returns.
- Sharpe & Sortino Ratios: excess return over risk-free rate, scaled by total volatility (Sharpe) or downside deviation only (Sortino).
- Value-at-Risk (VaR) & CVaR: computed at the 95% confidence level using historical daily return distributions per scheme; CVaR captures the average loss beyond the VaR threshold (tail risk).
- Herfindahl-Hirschman Index (HHI): sum of squared market-share percentages, used to quantify concentration in both sector holdings and fund-house AUM.
- Cohort Retention: SIP investors grouped by first-transaction month; retention tracked as the percentage of each cohort still transacting in subsequent months.
- Composite Fund Scorecard: z-score normalized blend of 3-year return, Sharpe, Sortino, Alpha, volatility (inverted), and max drawdown (inverted), rescaled to a 0–100 score.

4. Data Cleaning & Preparation

Key data quality issues identified and resolved during the pipeline build:

- Dynamic column-name detection to handle inconsistent headers across source files.
- Windows encoding fixes (UTF-8 BOM handling) for CSV exports.
- Forward-filled missing NAV values on non-trading days to maintain a continuous time series.
- Expense ratios cross-validated against SEBI TER caps by category.
- Chart palette indexing corrected for consistent color-coding across all Power BI visuals.
- AUM aggregation bug fixed: original dashboard summed AUM across all quarterly snapshots within a calendar year (e.g., adding Mar-31, Sep-30, and Dec-31 values together), inflating annual AUM by up to 3×. Corrected by using only the latest snapshot date within each year for year-over-year comparisons.

5. Exploratory Data Analysis

This section highlights the top insights from EDA across NAV trends, AUM growth, SIP inflows, and investor demographics.

5.1 Industry AUM Growth

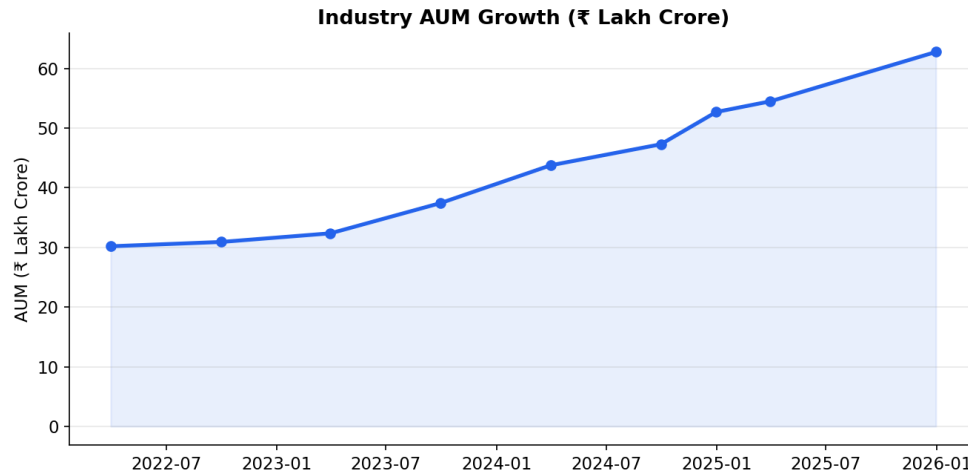


Figure 5.1 — Total industry AUM grew from ₹30.18 lakh crore to ₹62.74 lakh crore between Mar 2022 and Dec 2025 (~21.5% annualized).

AUM growth has been consistent across every snapshot with no observed contraction once the aggregation bug (Section 4) was corrected — the earlier draft dashboard's apparent 2025 decline was an artifact of double-counting, not a real trend.

5.2 AUM by Fund House

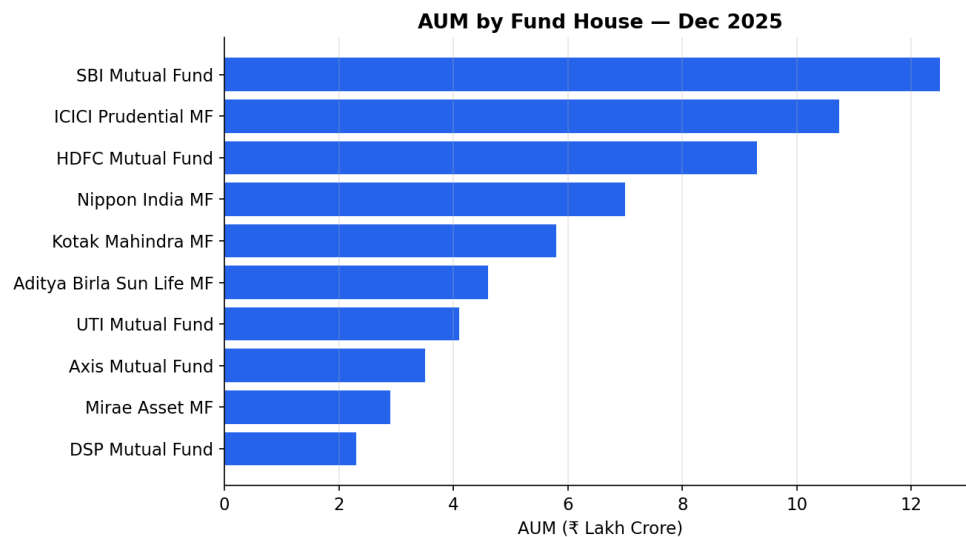


Figure 5.2 — SBI Mutual Fund leads AUM share (19.9%), followed by ICICI Prudential (17.1%) and HDFC (14.8%).

The fund-house HHI of 1,282 falls in the 'moderately concentrated' band (1,000–1,800 per standard antitrust thresholds), meaning the top three players hold just over half the market (51.9%) while seven other AMCs compete for the remainder.

5.3 SIP Inflow Trend

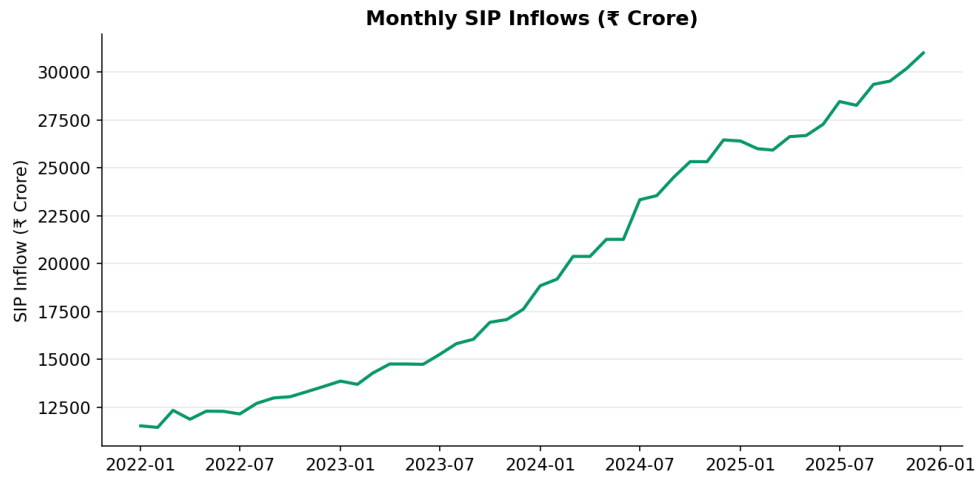


Figure 5.3 — Monthly SIP inflows show a strong upward trend, growing from ₹1,49,437 Cr (2022 total) to ₹3,35,740 Cr (2025 total).

5.4 Category-wise Net Inflows

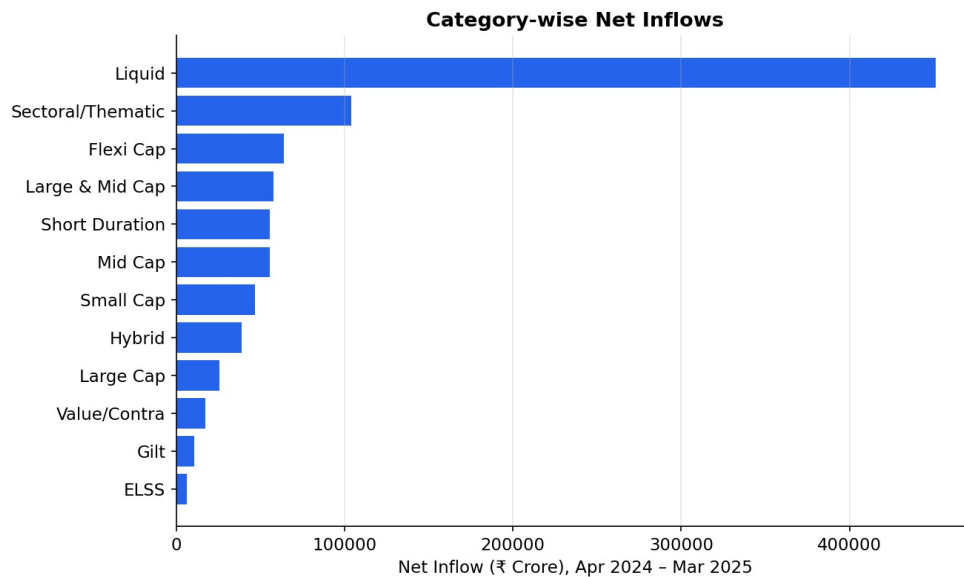


Figure 5.4 — Liquid funds dominate net inflows (₹4,51,275 Cr), reflecting short-term cash parking; Sectoral/Thematic funds rank second (₹1,03,829 Cr).

5.5 Investor Demographics

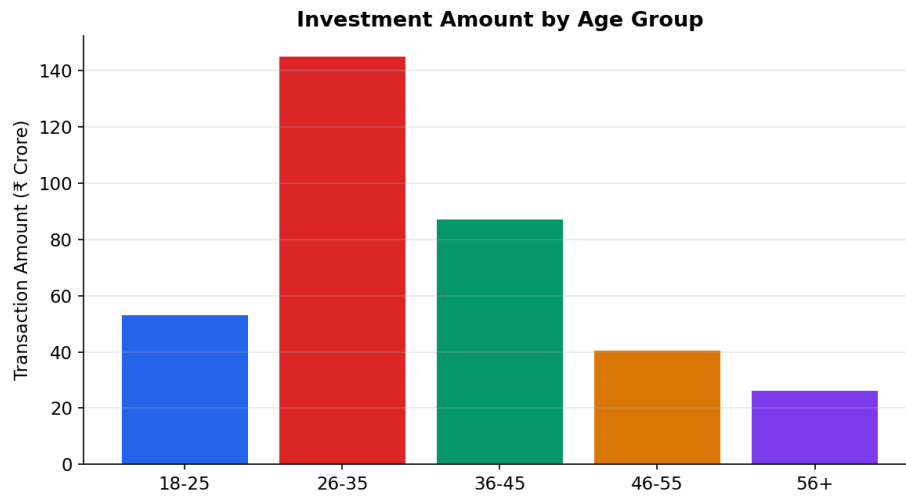


Figure 5.5 — Investors aged 26–35 contribute the highest transaction value (₹145.2 Cr), consistent with peak early-career wealth accumulation.

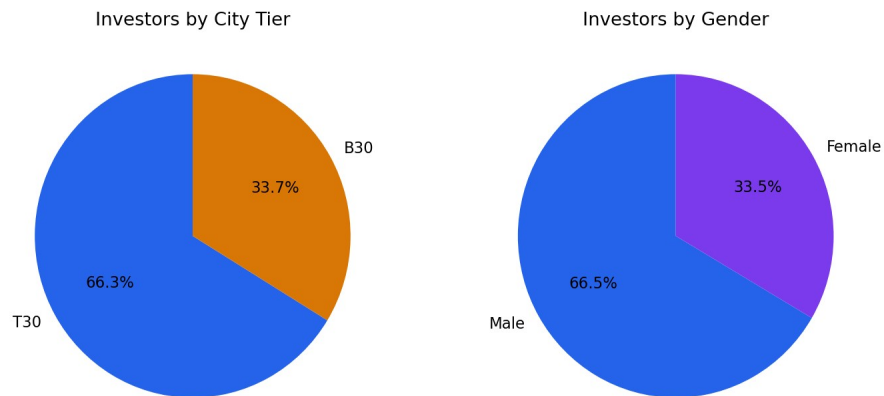


Figure 5.6 — 66% of investors are from Tier-30 (metro/large) cities; 66.5% of investors are male vs. 33.5% female.

5.6 Portfolio Sector Allocation

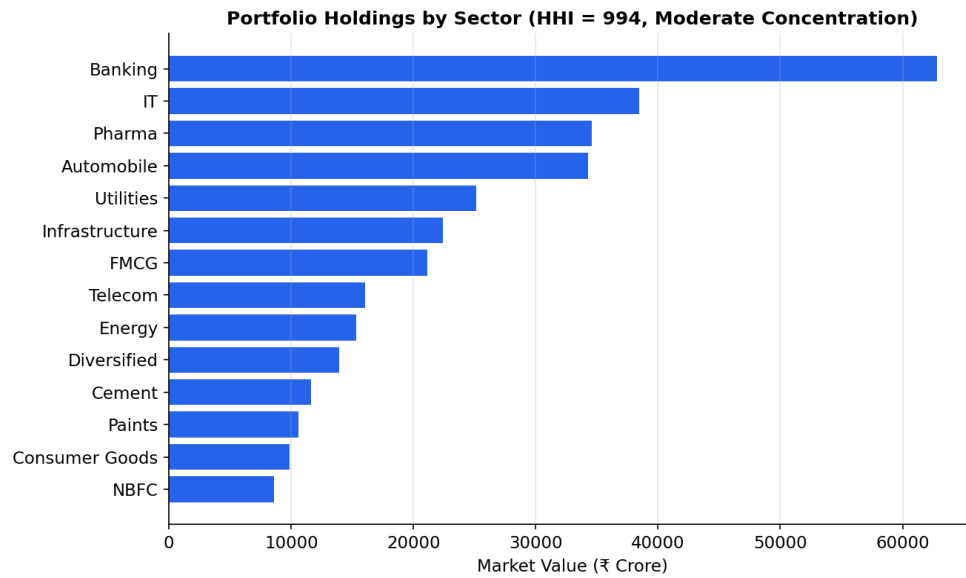


Figure 5.7 — Banking is the largest sector allocation (19.3%), followed by IT and Pharma; overall sector HHI = 994 (moderately diversified).

5.7 Industry Folio Growth

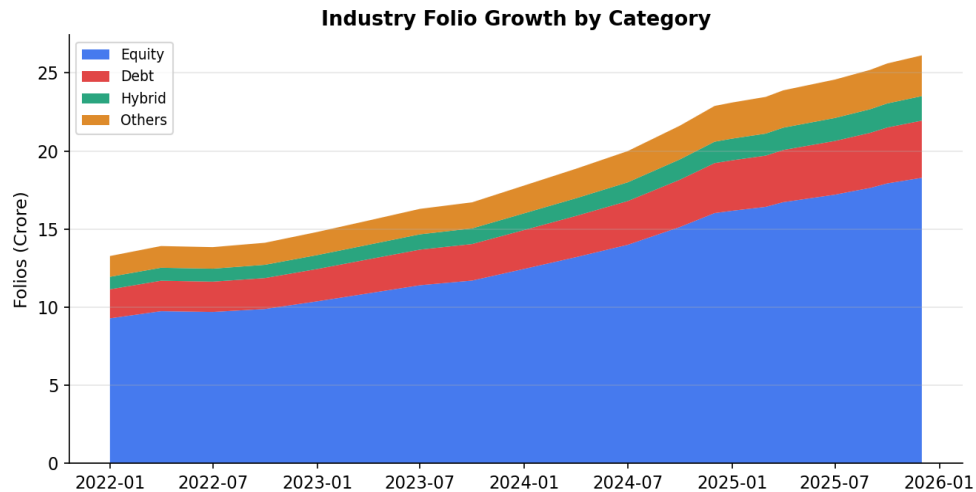


Figure 5.8 — Equity folios remain the dominant contributor to total industry folio growth across 2022–2025.

6. Performance Analytics

6.1 Alpha vs. Beta

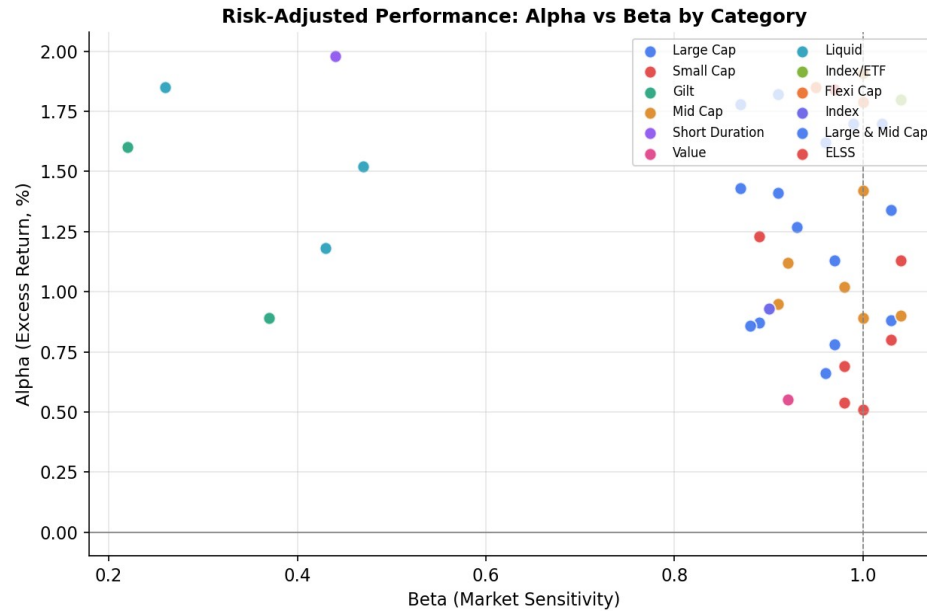


Figure 6.1 — Most schemes cluster near Beta ≈ 1 (market-matching risk) with positive Alpha, indicating modest consistent outperformance; Liquid and Gilt funds show low Beta as expected for their debt-oriented mandates.

Across all 40 schemes, the average 3-year annualized return is 14.09% with an average Alpha of 1.25% — meaning the typical fund in this sample outperformed its benchmark modestly after accounting for risk exposure.

6.2 Composite Fund Scorecard — Top 10

Each scheme is scored on a blended, z-score normalized basis across 3-year return, Sharpe Ratio, Sortino Ratio, Alpha, volatility, and Max Drawdown, rescaled to 0–100.

Scheme	Category	3Y Return %	Sharpe	Alpha	Score /100
ICICI Pru Liquid Fund	Liquid	7.68	7.68	1.85	100.0
Kotak Liquid Fund	Liquid	6.18	6.18	1.52	83.6
ABSL Liquid Fund	Liquid	5.14	5.14	1.18	70.0
HDFC Short Term Debt Fund	Short Duration	7.37	1.84	1.98	47.4
SBI Magnum Gilt Fund	Gilt	6.07	1.52	1.60	39.4
UTI Flexi Cap Fund	Flexi Cap	15.34	0.96	1.79	31.6
Kotak Flexicap Fund	Flexi Cap	15.65	0.98	1.85	28.1
Nippon India Gilt	Gilt	5.31	1.33	0.89	28.0

Scheme	Category	3Y Return %	Sharpe	Alpha	Score /100
Securities Fund					
Mirae Asset Large Cap Fund	Large Cap	14.81	1.06	1.62	28.0
Kotak Emerging Equity Fund	Mid Cap	18.23	0.96	1.91	26.9

Table 6.1 — Liquid and short-duration debt funds top the composite scorecard due to their very high Sharpe ratios (low volatility relative to return), not because they deliver the highest absolute returns.

Note for interpretation: this scorecard rewards risk-adjusted consistency, so low-volatility debt/liquid funds naturally score higher than high-return equity funds. For investors seeking growth rather than capital preservation, the scorecard should be filtered by category (e.g., ranking only within Equity or only within Mid/Small Cap) rather than compared across the full universe.

7. Risk Analytics

7.1 Value-at-Risk (VaR) & Conditional VaR (CVaR)

VaR and CVaR were computed at the 95% confidence level using each scheme's historical daily return distribution. VaR estimates the worst expected daily loss under normal conditions; CVaR (Expected Shortfall) estimates the average loss in the worst 5% of trading days — a more conservative tail-risk measure.

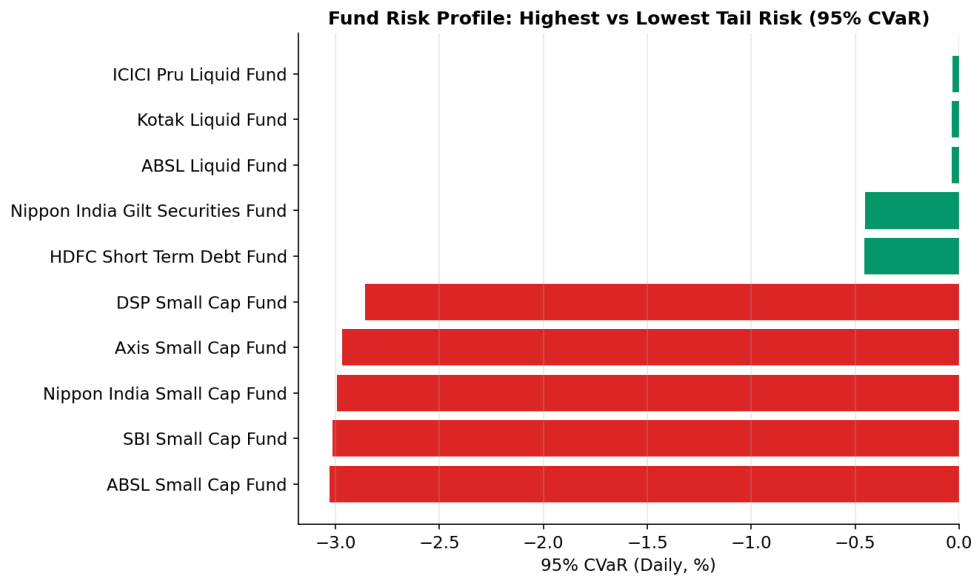


Figure 7.1 — Small Cap funds show the deepest tail risk (CVaR $\approx$ -3.0% daily), while Liquid funds show near-zero tail risk (CVaR $\approx$ -0.03%), confirming the expected risk hierarchy across categories.

Across the sample, the average annualized volatility is 12.6%, ranging from 0.46% (Liquid funds) to 21.8% (Small Cap funds) — a roughly 47x spread that underlines how critical category selection is to an investor's risk exposure, independent of fund selection within a category.

7.2 Sector Concentration Risk (HHI)

Portfolio holdings across all 40 schemes were aggregated by sector to assess concentration risk. The overall sector HHI is 994, placing the combined portfolio in the 'unconcentrated' to 'moderately concentrated' boundary (HHI < 1,000 is generally considered a competitive/diversified market by standard antitrust benchmarks).

Sector	Market Value (₹ Cr)	Share %
Banking	62,840.29	19.34%
IT	38,477.11	11.84%
Pharma	34,606.10	10.65%
Automobile	34,296.97	10.56%
Utilities	25,108.63	7.73%
Infrastructure	22,433.39	6.91%
FMCG	21,151.15	6.51%
Telecom	16,051.45	4.94%

Table 7.1 — Top 8 sectors by aggregated portfolio market value across all 40 schemes (Dec 2025 snapshot).

8. Investor Behaviour Analysis

8.1 SIP Cohort Retention

SIP investors were grouped into cohorts by the month of their first SIP transaction. Retention tracks what percentage of each cohort continued transacting in subsequent months.

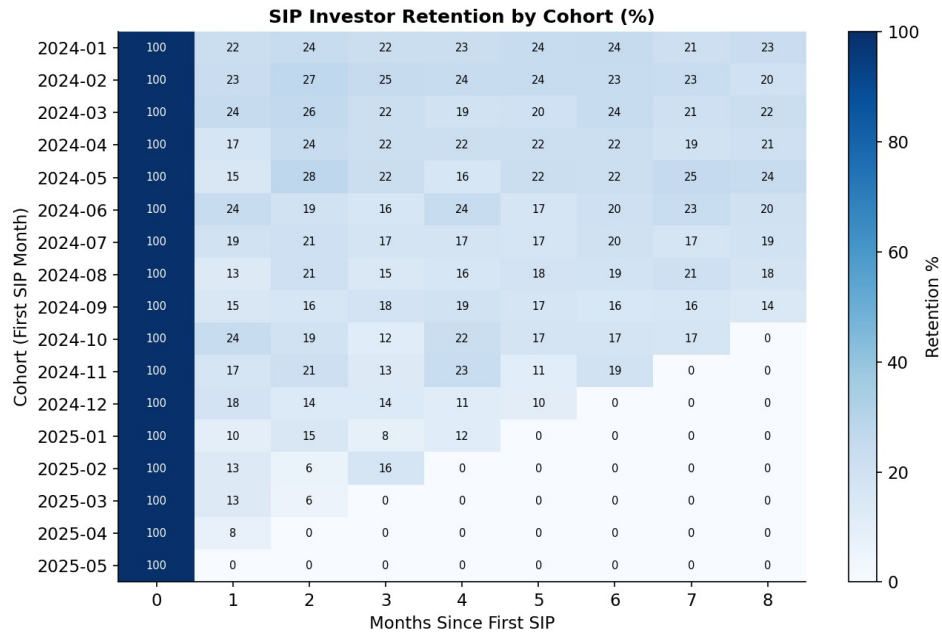


Figure 8.1 — Retention drops sharply after month 1 (100% → ~16%) then stabilizes near 13–17% through month 5, suggesting a bimodal investor base: a core of consistent long-term SIP holders and a larger group of one-off or irregular contributors.

This pattern is common in retail SIP data where monthly transaction timing doesn't align perfectly month-to-month (e.g., a SIP debited on the 3rd one month and the 1st the next may appear as a 'gap' in a naive month-over-month count). This is worth flagging as a modeling caveat: true retention is likely understated by this simple cohort method and would benefit from a rolling 60-90 day activity window in a future iteration.

8.2 Demographic & Transaction Summary

Metric	Value
Unique Investors	5,000
Total Transactions	32,778
Total Transaction Value	₹352.16 Cr
SIP Transactions	19,716 (60.1%)
Lumpsum Transactions	8,095 (24.7%)
Redemptions	4,967 (15.2%)
KYC Verified	30,146 (92.0%)
Top Contributing Age Group	26–35 (₹145.2 Cr)

9. Recommendations

- Prioritize the top-quartile equity funds by within-category composite score (not the cross-category scorecard) when building an equity-focused fund recommender, since the blended score structurally favors low-volatility debt/liquid funds.
- Flag Small Cap and Mid Cap schemes with CVaR beyond -2.5% for enhanced risk disclosure to conservative investors during onboarding.
- Target retention campaigns at the month-2 mark for SIP investors, where the steepest cohort drop-off occurs, using a rolling activity window rather than strict calendar-month matching to avoid overstating churn.
- Monitor sector HHI concentration (currently 994) as new fund launches are added — Banking's 19.3% share is not yet a systemic risk but warrants a rebalancing check if it exceeds ~25%.
- Correct the AUM aggregation logic in all downstream Power BI reports (use latest snapshot per period, not a sum) to avoid repeating the 3× AUM inflation identified in this project.

10. Conclusion

This capstone successfully delivers an end-to-end mutual fund analytics platform — from raw NAV and transaction data through cleaned star-schema tables, performance and risk analytics, investor behaviour modeling, and an interactive Power BI dashboard. The analysis confirms healthy, real (non-inflated) AUM growth of ~21.5% annualized, identifies moderate market and sector concentration, and surfaces actionable investor retention and risk-disclosure insights. The corrected AUM aggregation logic and composite Fund Scorecard are the two most directly reusable outputs for Bluestock Fintech's ongoing investment research workflow.

Appendix: Repository & Data Sources

GitHub Repository: github.com/Tejaswini-0625/bluestock_mf_capstone

Data Source: mfapi.in (NAV history), SEBI category/expense-ratio references, synthetic investor transaction & holdings data modeled on realistic Indian mutual fund industry distributions.